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PRIYADARSHINI COLLEGE OF ENGINEERING



DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING

HACKATHON 2026

Idea Submission Template — Code. Create. Innovate.

Team Name:

Team Raksha

Problem Statement No:

2

Problem Statement Title:

Improving Safety at Large Public Events

Team Leader Name:

[Your Name]

IDEA TITLE

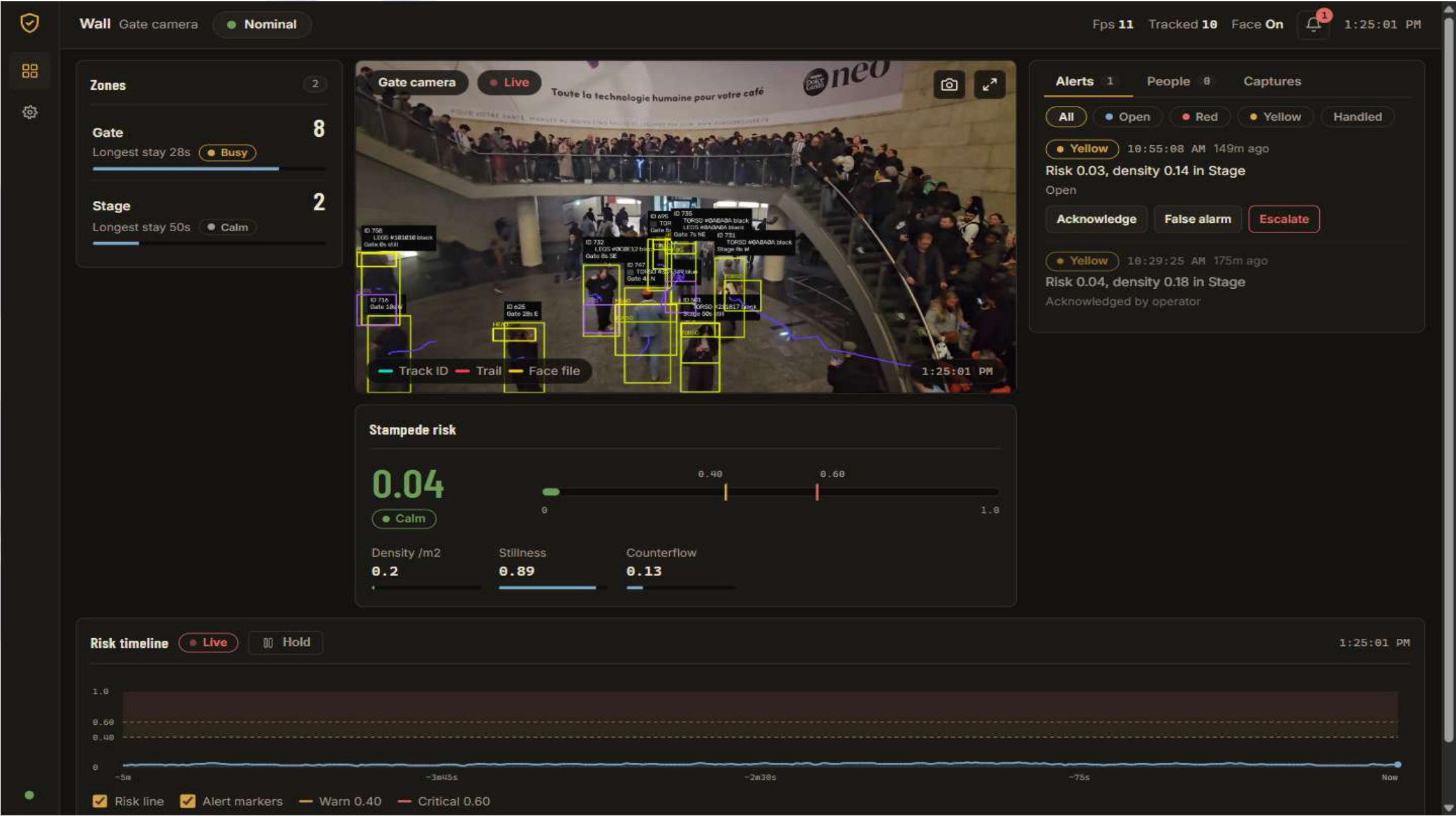
Know the crowd. Know the person. Stop the surge.

Live tracking plus instant person files.

Surge warnings before the crush.

0.4s <1s 15

face to file pop yellow to guard fps live wall



Above: the live wall on real station footage.

TECHNICAL APPROACH



Quality gate

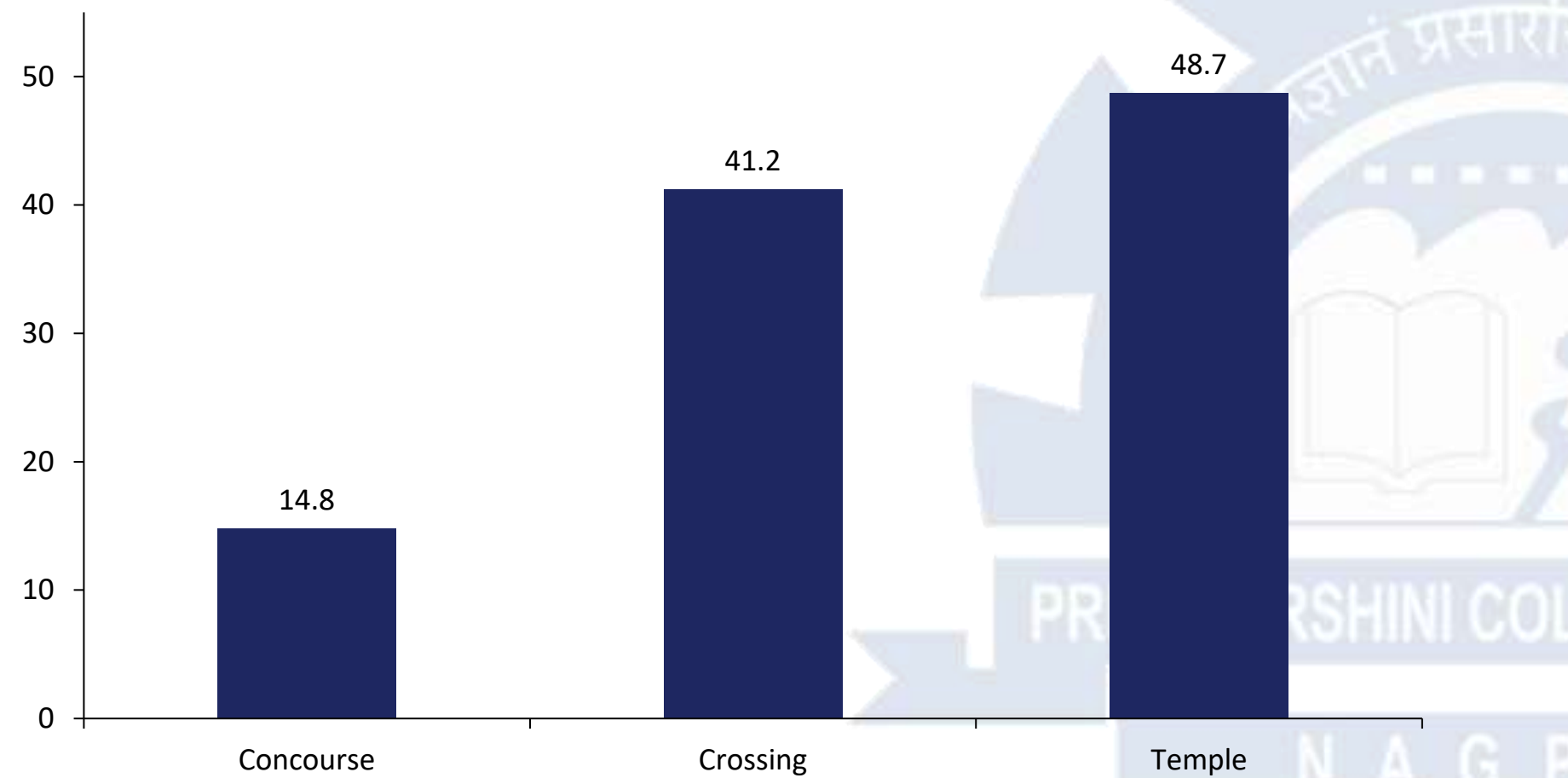
Only sharp frontal faces enroll. No junk matches.

Measured

Peak 18 tracked at once, 850 IDs over 2 min of concourse.

FEASIBILITY AND VIABILITY

Tracking speed, frames per second



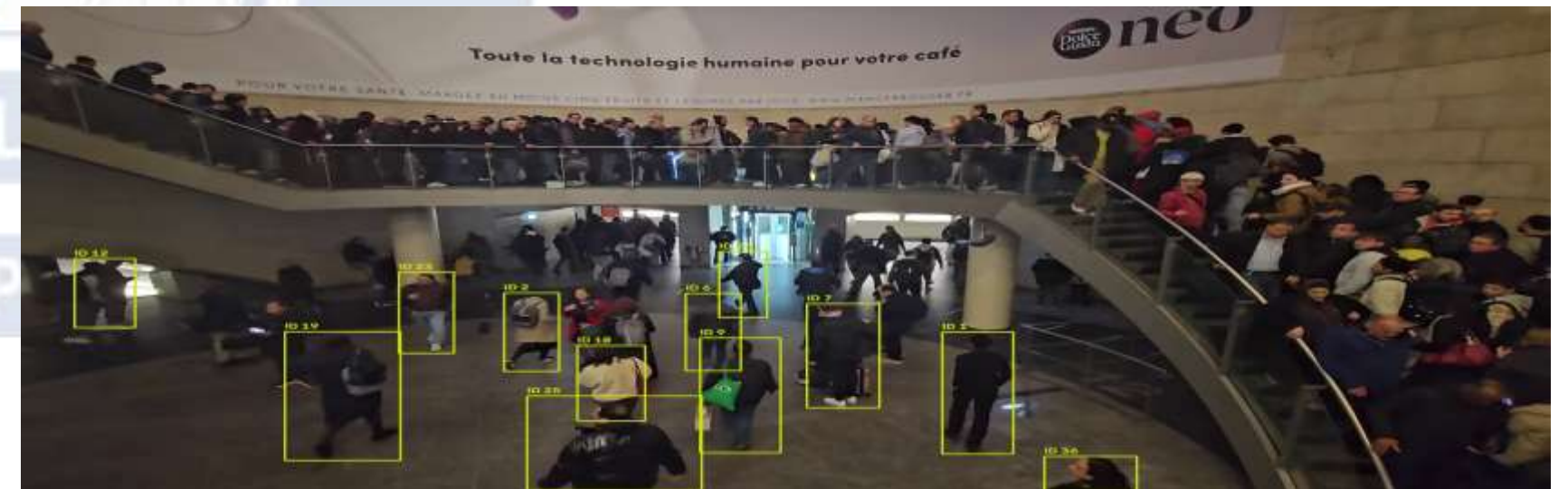
YOLOv8s plus ByteTrack on RTX 4050 laptop. No extra hardware.

61 ms

face match on GPU, was 438 ms on CPU

1.2+

risk score where RED fired on dense test



IMPACT AND BENEFITS



Before: 50 screens, one tired operator.

After: one wall, zone-wise push to guards.

Organizers

Bottleneck flagged 60s before humans see it

Police

Photo search under 2s, dossier plus trail in one click

Public

Safer exits, faster help, blurred storage

RESEARCH AND REFERENCES

YOLOv8 plus ByteTrack, 2025-26 studies: best speed balance; ByteTrack 14 percent faster than BoT-SORT.

ArcFace buffalo plus Faiss: 99.5 percent LFW, cosine 0.40, vote smoothing steadies matches.

Stampede models: AURORA hybrid, FTLE bottlenecks, Delhi pilot cut response 37 percent.

India context: Hathras, Kumbh, station crushes come from density plus blocked exits.

Blurred storage

faces blurred on disk

7-day purge

auto delete, no backlog

Audit log

hash-chained actions

No ID linkage

opt-in gallery only